DSA Assignment # 2

Name: Muhammad Ahmed
Roll No: SP24-BSE-066
Section: B

"Algorithm for hospital room (ER) managment"

① Patient class:

First of all create a patient class. It contain two variables type and id. and make it constructor

→ Make a method isType(); to check if patient is critical or 2nd

→ Constructor → Patient(String type, String id){
this.type = type;
this.id = id;
}

② Priority Queue class:

Then creat Priority Queeu class containting:

Patient patient[];
int size;
int rean;
PriorityQueue(int size){
patient = new Patient[size];
rean = -1
this.size = size;
}

→ isEmpty() and isFull() methods:
Create these method to check if Queue is full or empty.

→ insert method:

```
public void insert(Patient p) {
    CHECK ISFULL()
    patient[++rear]=p;
```

→ highPriority method:

```
for(int i=0; i<=rear; i++){
    if(patient[i].isType()){
        return i;
    }
}
```

→ remove method:

```
public Patient remove(){
    CHECK IF IS EMPTY()
    int highpriority=highPriority();
    Patient p = patient[highPriority];
    for (int i=highPriority; i<=rear; i++){
        patient[i]=patient[i+1];
    }
    rear--;
    return p;
}
```

→ remore by ID (String Id){

```
CHeck IF ISEMPTY()
for (int i=0 ; i≤rean; i++){
        if (patient [i].id.equals (id){
        patient [i] = patient [i+1];
    rean --;
    }
  return
}
```

→ display method:

Then create display method to display allremaining patients

```
for (int i=0; i≤rean; i++){
Sout ("Patient-type": TYPE+ "Patient-ID"+ ID);
}
```

③ ER Managment System class:

Here we call Priority Queue are call the given methods

① addPatient (String type, String Id) → call insert method
② treatPatient ( ) → call remove method
③ displayPatients () → call display method
④ declareEmergency (); → call remove by Id method